

*Version en français disponible sur demande

Education

- Expected 2027 **Ph.D. in Physics**, Université de Montréal.
Oct-Dec 2026: Balzan Research Fellow at New College, University of Oxford
Jul-Sept 2025: Mitacs Globalink visiting researcher at University of Oxford
Supervised by Yashar Hezaveh and Laurence Perreault-Levasseur
- 2022 **B.S. in Physics and Astronomy**, *Summa Cum Laude*, with honors, Graduate School Preparation Track, University of Pittsburgh.
- 2022 **B.A. in Russian**, *Summa Cum Laude*, University of Pittsburgh.

Research Profile

My current research combines galaxy dynamics, strong gravitational lensing, and radio interferometry to measure supermassive black hole masses in nearby and distant galaxies. I develop differentiable astrophysical simulators and Bayesian inference methods that fit ALMA observations directly in visibility space. I also develop uncertainty-aware machine learning methods for interferometric reconstruction, with a broader interest in understanding and controlling how modelling assumptions affect astrophysical measurements. My earlier work includes developing a new galaxy cluster-finding algorithm that incorporates redshift uncertainties from photometric surveys such as the DESI Legacy Survey.

Scholarships and Fellowships

- 2026 **Balzan Research Fellow at New College, University of Oxford**, Competitive award funding my visiting researcher role at the University of Oxford for a term (hosted as the equivalent to a Visiting Junior Research Fellow).
- 2025 **Mitacs Globalink Research Award**, Competitive grant to support my visiting researcher role at the University of Oxford for the summer.
- 2024 **Bourse Gilles Beaudet**, Competitive scholarship award for PhD students in the physics department of the University of Montreal.
- 2024 **Bourse du passage accéléré UdeM**, Competitive scholarship award for successfully transitioning from a masters to PhD on the accelerated track.
- 2022-Present **Bourse d'exemption UdeM**, Competitive international student scholarship, covers tuition fees.
- 2022 **Bourse d'excellence du centenaire**, Scholarship awarded to a top incoming graduate student in the physics department at University of Montreal.
- 2022 **Bourse d'excellence des ESP**, Scholarship awarded to a top incoming graduate student in the physics department at University of Montreal.
- 2020, 2022 **NASA Pennsylvania Space Grant Consortium Research Award**, Undergraduate summer research scholarship.
- 2021 **NSF (National Science Foundation) Research Experiences for Undergraduates (REU) at NRAO**, Undergraduate summer research scholarship to fund a research internship at the National Radio Astronomy Observatory (NRAO).
- 2018-22 **University of Pittsburgh Full Tuition Scholarship**.

Accepted Telescope Observing Proposals

5 accepted ALMA proposals (58.4 h); led the mock-observation simulations and visualizations for all proposals.

- 2026 **Co-I, Atacama Large Millimeter/submillimeter Array (ALMA)**, *33.9 hours*, The first direct dynamical constraint on the supermassive black hole mass - stellar mass relation at $z = 4$ (PI: Hengyue Zhang).
- 2026 **Co-I, Atacama Large Millimeter/submillimeter Array (ALMA)**, *9.5 hours*, The final building block for the first dynamical mass measurement of an inactive supermassive black hole at $z = 4$ (PI: Hengyue Zhang).
- 2025 **Co-I, Atacama Large Millimeter/submillimeter Array (ALMA)**, *5.3 hours*, A gas-dynamical SMBH mass measurement at $z = 4.23$: probing the sphere of influence of SPT0113 at 50 pc resolution (PI: Jacob Elford).
- 2025 **Co-I, Atacama Large Millimeter/submillimeter Array (ALMA)**, *1.1 hours*, First dynamical supermassive black hole mass measurement at $z \sim 4$: resolving the lensed sphere of influence of ID141 (PI: Hengyue Zhang).
- 2024 **Co-I, Atacama Large Millimeter/submillimeter Array (ALMA)**, *8.6 hours*, First dynamical supermassive black hole mass measurement at $z \sim 4$: resolving the sphere of influence of a $z=4.24$ galaxy (PI: Hengyue Zhang).

Open Source Software

Projects where I am the creator and lead developer are listed in bold.

- 1. **SuperMAGE: Superb masses from gas kinematics**. A differentiable, modular gas dynamics simulator for galaxies.
Tech Stack: PyTorch, Caskade, VisCube, astropy, numpy, scipy
<https://github.com/mjyb16/supermage>
- 2. **VisCube: Visibility-space gridder for spectral cubes**. Pre-processing software for radio interferometry data gridding and uncertainty quantification in the visibility plane.
Tech stack: numpy, scipy, casa, XRADIO
<https://github.com/mjyb16/viscube>
- 3. **Caustics**: A gravitational lensing simulator for the machine learning era.
Tech Stack: PyTorch, Caskade
<https://github.com/Ciela-Institute/caustics>

Research Mentoring

- 2023-Present **Bayesian Imaging for Radio Interferometry with Score-Based Priors**, University of Montreal.
Main supervisor of undergraduate student Noé Dia (resulting in a peer-reviewed workshop paper and a journal manuscript submitted to ApJ).

Professional Service and Teaching

- 2026 **Peer reviewer**, *NeurIPS (Neural Information Processing Systems) Sim2Science Workshop*, Reviewed two papers submitted to the workshop.
- 2026 **Peer reviewer**, *Machine Learning and the Physical Sciences Workshop*, Reviewed three papers submitted to the workshop.
- 2025-26 **Peer reviewer**, *Astronomy & Astrophysics (A&A)*, Reviewed a submitted paper.
- 2025 **Peer reviewer**, *ICML (International Conference on Machine Learning) ML4Astro Workshop*, Reviewed two papers submitted to the workshop.
- 2024 **New student mentor**, *Ciela Institute*, Taught incoming graduate students how to set up and use Digital Research Alliance of Canada (Compute Canada) clusters.

- 2023 **Organizing committee member and teaching instructor**, *Astromatic Hackathon at Ciela Institute*, University of Montreal.

Presentations (selected)

- 2026 **Talk**, "**Dynamical supermassive black hole masses across cosmic time**", *American Museum of Natural History (Astronomy Seminar)*, 17 September 2026.
- 2026 **Talk**, "**Dynamical supermassive black hole masses across cosmic time**", *Columbia University (Astronomy Seminar)*, 14 September 2026.
- 2026 **Talk**, "**Dynamical supermassive black hole masses across cosmic time**", *PASSAGES group meeting*, 8 September 2026.
- 2026 **Talk**, "**Efficient uncertainty-aware learning on interferometric data with VisCube**", *2nd Annual Cosmic Horizons Conference (hosted by the National Radio Astronomy Observatory)*, 14 July 2026.
- 2026 **Poster presentation**, "**Dynamical SMBH Mass Measurement in Obscured Galaxies via Joint SMBH-Stellar Mass Modeling**", *Supermassive Black Holes and Blue Notes Conference*, 29 June - 3 July 2026.
- 2026 **Talk**, "**Dynamical SMBH Mass Measurement in Obscured Galaxies via Joint SMBH-Stellar Mass Modeling**", *Canadian Astronomical Society (CASCA) Meeting*, 11 June 2026.
- 2026 **Talk (in French)**, "**Mesure directe de la masse des trous noirs supermassifs dans les galaxies obscurcies par moyen de modélisation neuronale de la distribution stellaire**", *Association canadienne-française pour l'avancement des sciences (ACFAS)*, 11 May 2026.
- 2025 **Poster presentation**, "**Neural Deprojection of Galaxy Stellar Mass Profiles**", *Neural Information Processing Systems conference, Machine Learning for Physical Sciences workshop*, 06 December 2025.
- 2025 **Talk**, "**Supermassive black hole mass measurement across cosmic time using visibility-space modelling of molecular gas dynamics and flexible stellar mass parametrizations**", *Department of Physics, University of Oxford*, 26 September 2025.
- 2025 **Talk**, "**Supermassive black hole mass measurement across cosmic time using visibility-space modelling of molecular gas dynamics and flexible stellar mass parametrizations**", *School of Physics and Astronomy, Cardiff University*, 4 September 2025.
- 2024 **Talk**, "**Score-based Bayesian Imaging for Interferometry**", *Spatio-spectral Modeling of Interferometric Data Workshop, National Radio Astronomy Observatory*, 29 May 2024.
- 2023 **Poster Presentation**, "**CluMPR: A new galaxy cluster-finding algorithm**", *Statistical Challenges in Modern Astronomy VIII*, 13 June 2023.
- 2022 **Poster Presentation**, "**CluMPR: A new galaxy cluster-finding algorithm**", *240th Meeting of the AAS*, 13 June 2022.
- 2021 **Talk**, "**SiRIUS: Simulation of Radio Interferometry from Unique Sources**", *NRAO Summer Student Symposium at Green Bank Observatory*, 2 August 2021.

Professional Development (selected)

- 2025 **NRAO ALMA Cycle 12 Proposal Preparation workshop**, gave invited talk.
- 2023 **Statistical Challenges in Modern Astronomy VIII conference and Summer School in Statistics for Astronomers XVIII**, gave poster presentation.
- 2023 **CRAQ (Center for Research in Astrophysics of Quebec) conference**, gave flash talk.
- 2022 **NRAO 18th Synthesis Imaging Workshop**.

Community Service

- 2026 **Public Outreach Guide**, *Eureka Science Festival (Montreal)*, Organized and supervised hands-on astronomy activities for children and their families at a bilingual science festival.
- 2026 **High School Student Career Day Mentor**, *University of Montreal*, French-language Q&A with high school students visiting the department.
- 2026 **Public Lecturer**, *Astronomy on Tap (Montreal)*, Gave a general-audience talk in French and helped with event organization.
- 2025 **Public lecturer**, *Club des Astronomes Amateurs de Rosemère*, French-language outreach talk on cosmology.
- 2023-2025 **Organizing committee**, *Student Symphony Orchestra (OSEUM)*, University of Montreal.
- 2019-2020 **Public Outreach Volunteer/Guide**, *Buhl Planetarium at the Carnegie Science Center*.
- 2017-2018 **Public Outreach Volunteer/Guide**, *Observatory at Turner Farm, Great Falls, VA*.

Computer Skills

Programming and development tools: Python, bash/SLURM, git, LaTeX, XML/HTML/CSS, Mathematica, Java, Swift, SQL

Python Libraries: PyTorch, Dask, KeOps, Numba, Scikit-learn, Pandas, numpy/scipy, astropy/photutils, Matplotlib, astroalign

Software: CASA (Common Astronomy Software Applications) radio astronomy data reduction and analysis suite

Languages

*Self-assessed CEFR level, not formally tested.

English	Native Language
Russian	C2 (heritage/native)
French	B2/C1

Research Interests

Strong gravitational lensing, galaxy dynamics, supermassive black holes, radio interferometry, statistics, Bayesian inference, neural networks/machine learning, galaxy clusters/large scale structure, multiwavelength galaxy surveys, measurement of cosmological parameters, high-performance computing

Referees

Martin Bureau, Professor, *Department of Physics, University of Oxford*, martin.bureau@physics.ox.ac.uk.

Yashar Hezaveh, Professor, *Department of Physics, University of Montreal*, yashar.hezaveh@umontreal.ca.

Laurence Perreault-Levasseur, Professor, *Department of Physics, University of Montreal*, laurence.perreault.levasseur@umontreal.ca.

Jeffrey Newman, Professor, *Department of Physics and Astronomy, University of Pittsburgh*, janewman@pitt.edu.

Publications

Since I work at the interface of both the astronomy and machine learning communities, this publication list contains both astronomy journal articles and machine learning workshop papers. The machine learning workshop papers listed below underwent double-blind peer review of the submitted manuscripts, rather than selection based solely on presentation abstracts.

Publications are listed in reverse chronological order. Titles of first- and joint-first-author papers are in bold. Shared first-authorship is indicated with a dagger ([†]). Papers led by an undergraduate student under my direct supervision are indicated with an asterisk (*). Note that I publish under the name M. J. Yantovski-Barth rather than my legal name (Michael Barth).

8. [†]**GLAMOR II. A gas-dynamical supermassive black hole mass measurement in a gravitationally-lensed galaxy at $z=4.2$**
In prep.
[†]Hengyue Zhang, [†]**M. J. Yantovski-Barth**, et al.
7. [†]**GLAMOR I. A supermassive black hole mass measurement using visibility-space modelling of molecular gas dynamics and imaging-free stellar mass parametrizations**
In collaboration-wide review.
[†]**M. J. Yantovski-Barth**, [†]Hengyue Zhang, et al.
6. **When precision exceeds fidelity: systematic error quantification for inference with misspecified simulators**
Accepted to the Sim2Science Workshop at 2026 NeurIPS conference, 12 Dec 2026.
M. J. Yantovski-Barth, Hengyue Zhang, et al.
5. *IRIS: A Bayesian approach for image reconstruction in radio interferometry with expressive score-based priors
Submitted to Astrophysical Journal.
*Noé Dia, ***M. J. Yantovski-Barth**, et al.
<https://ui.adsabs.harvard.edu/abs/2025arXiv250102473D/abstract>
4. **Neural Deprojection of Galaxy Stellar Mass Profiles**
ML4PS Workshop at 2025 NeurIPS conference, 6 Dec 2025.
M. J. Yantovski-Barth, Hengyue Zhang, et al.
<https://ui.adsabs.harvard.edu/abs/2025arXiv251120746Y/abstract>
3. Caustics: A Python Package for Accelerated Strong Gravitational Lensing Simulations
Journal of Open Source Software, 9(103), 7081, 22 November 2024.
Connor Stone, Alexandre Adam, Adam Coogan, **M. J. Yantovski-Barth**, et al.
<https://ui.adsabs.harvard.edu/abs/2024JOSS...9.7081S/abstract>
2. **The CluMPR Galaxy Cluster-Finding Algorithm and DESI Legacy Survey Galaxy Cluster Catalogue**
MNRAS, 531, 2, Jun 2024.
M. J. Yantovski-Barth, Jeffrey A. Newman, Biprateep Dey, et al.
<https://ui.adsabs.harvard.edu/abs/2024MNRAS.531.2285Y/abstract>
1. *Bayesian Imaging for Radio Interferometry with Score-Based Priors
ML4PS Workshop at 2023 NeurIPS conference, Dec 2023.
*Noé Dia, ***M. J. Yantovski-Barth**, et al.
<https://ui.adsabs.harvard.edu/abs/2023arXiv231118012D/abstract>